

Typographic-Attack Benchmark — Report

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This report covers Phase 1, Axis C of the project — a single semantic, model-agnostic attack (the typographic overlay) evaluated against the 7 ImageNet baselines listed in `config.yaml`. The poisoned dataset is generated once by `scripts/generate_datasets.py --typographic` and shared across all models; this report aggregates the per-model JSONs in this directory.

1. Setup

- **Hardware:** Tesla T4.
- **OS:** Linux-6.6.122+-x86_64-with-glibc2.35.
- **Libraries:** torch 2.10.0+cu128, torchvision 0.25.0+cu128, transformers 5.0.0, pillow 11.3.0.
- **Attack:** typographic overlay (`attacks/typographic.py`), white sticker with bold black text drawn near the top of each clean image. Config: `font_size_frac=0.12, position='top', padding_frac=0.04, opacity=1.0, jpeg_quality=90, text_form='lowercase_first_synonym'`.
- **Dataset:** the full 1000-image clean benchmark, one poisoned variant per clean image (`data/poisoned/typographic/`). Each poisoned image is stamped with the lowercase first synonym of a random *wrong* ImageNet class, picked deterministically with the global seed.
- **Global seed:** 42 (`config.yaml`). Set on `random`, `numpy`, `torch` at the start of every model.

2. Headline accuracy table

Model	Clean	Robust	Drop	Fooling rate	TASR
resnet50	0.781	0.768	0.013	0.036	0.001
vgg16	0.687	0.663	0.024	0.076	0.000
convnext_tiny	0.787	0.777	0.010	0.032	0.000
vit_b_16	0.793	0.773	0.020	0.039	0.001
swin_t	0.775	0.770	0.005	0.032	0.002
efficientnet_b0	0.743	0.722	0.021	0.055	0.001
clip_vit_b16	0.621	0.368	0.253	0.452	0.342

Columns: clean accuracy, robust accuracy (untargeted — did the model flip away from the true label?), accuracy drop (clean – robust), fooling rate (originally-correct images that flipped), and TASR — the **targeted-attack success rate**, the fraction of images where the model predicted the *exact* class named by the overlay text.

Machine-readable copy: `accuracy_table.csv`.

3. Sanity checks

- **Typographic overlay causes a non-zero accuracy drop on every model.**
- resnet50: clean **0.781** → robust **0.768** (drop **0.013**) → **PASS**.
- vgg16: clean **0.687** → robust **0.663** (drop **0.024**) → **PASS**.
- convnext_tiny: clean **0.787** → robust **0.777** (drop **0.010**) → **PASS**.
- vit_b_16: clean **0.793** → robust **0.773** (drop **0.020**) → **PASS**.
- swin_t: clean **0.775** → robust **0.770** (drop **0.005**) → **PASS**.
- efficientnet_b0: clean **0.743** → robust **0.722** (drop **0.021**) → **PASS**.
- clip_vit_b16: clean **0.621** → robust **0.368** (drop **0.253**) → **PASS**.
- **CLIP resists typographic more than the pure-classifier mean.** pure mean drop **0.016** vs CLIP drop **0.253** → **FAIL**.
- **Coverage:** 7 / 7 models reported. **PASS**.

4. Per-model analysis

Across the 7 evaluated models the typographic overlay caused an average accuracy drop of **0.05** absolute. The most affected model was clip_vit_b16, dropping from **0.62** clean to **0.37** robust (drop **0.25**, TASR **0.34**). clip_vit_b16 (zero-shot, language-grounded) recorded clean **0.62** → robust **0.37** (drop **0.25**, TASR **0.34**). Compare this directly with the pure-classifier average — that gap is the paper's language-grounding finding.

5. Figures

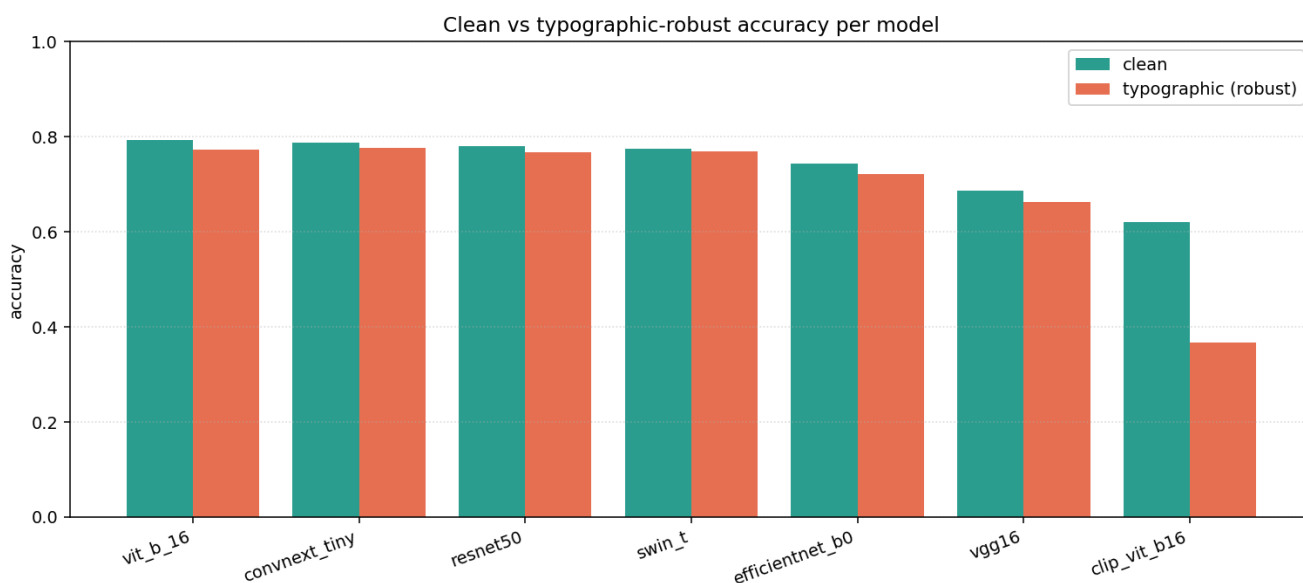


Figure 1 — Clean vs typographic-robust accuracy per model (sorted by clean acc).

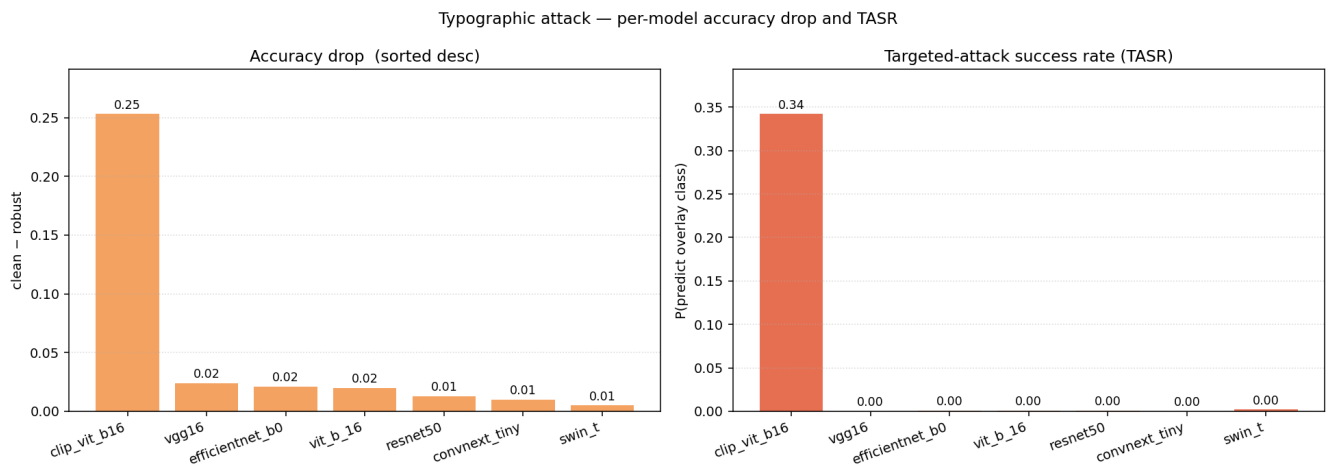


Figure 2 — Accuracy drop and targeted-attack success rate per model.

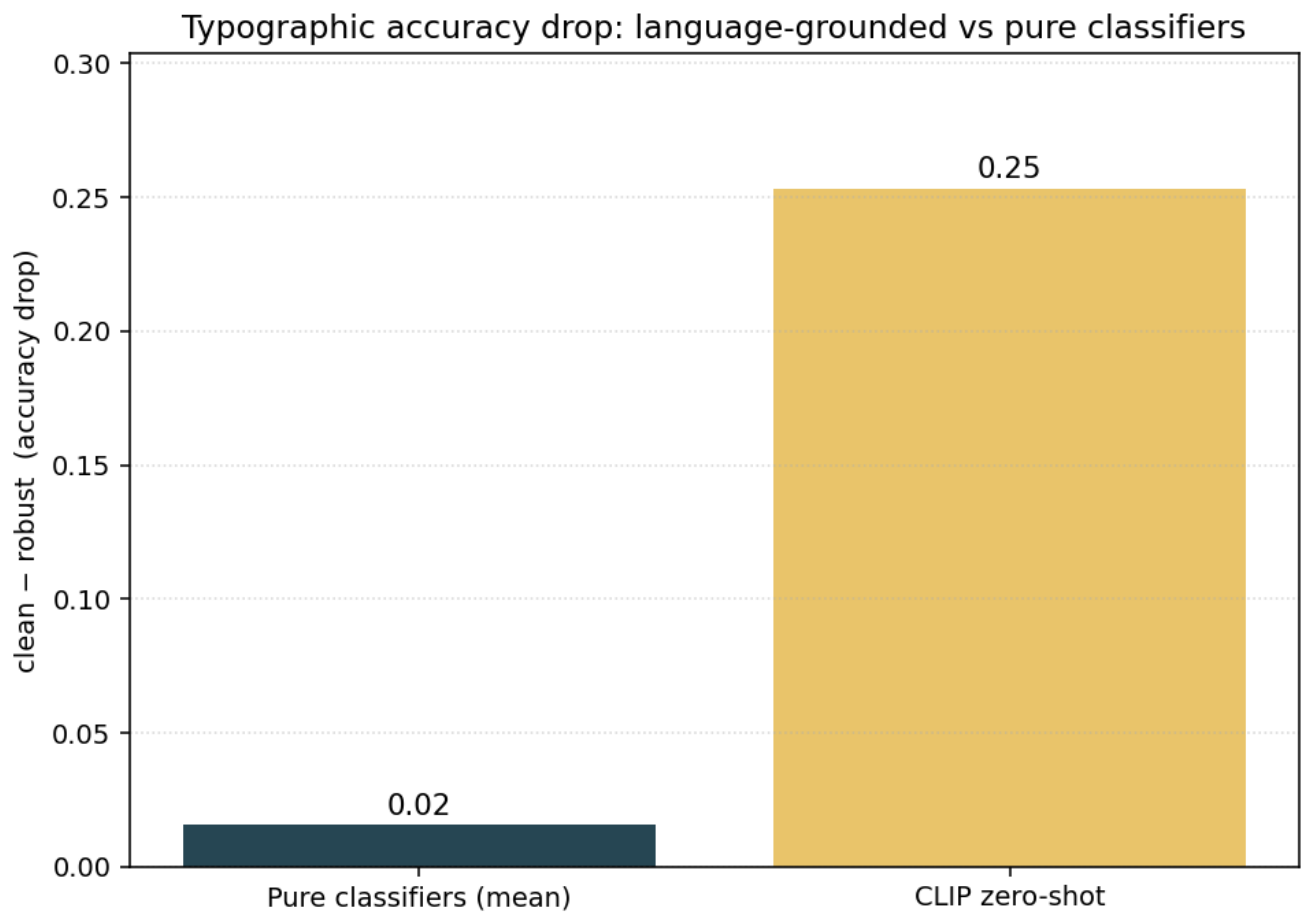


Figure 3 — Language-grounded (CLIP) vs pure-classifier mean drop.

Typographic examples (clean | poisoned)

clean: tench



typo overlay: "minibus"



clean: goldfish



typo overlay: "slug"



clean: great white shark



typo overlay: "european fire sala"



Figure 4 — Clean | poisoned examples (typographic overlay).

6. Interpretation

The typographic attack is the paper's headline semantic attack. Pure vision classifiers are expected to latch onto the rendered text token and predict the wrong class — the TASR column quantifies how often the model is fooled into the *exact* class named on the sticker, not just into any wrong class. A language-grounded model like CLIP classifies by matching against text descriptions of every class; the hypothesis (project brief §3.6, hook #1) is that this makes it resist text overlays compared to pure classifiers. Compare CLIP's drop to the pure-classifier average in the sanity-check block above.

7. Reproducibility footer

- **Wall-clock (this report build session):** 0.1 min.
- **Per-model cumulative compute (sum of per-cell wall_clock_s):** clip_vit_b16 0.5 min, convnext_tiny 0.3 min, efficientnet_b0 0.1 min, resnet50 0.2 min, swin_t 0.3 min, vgg16 0.2 min, vit_b_16 0.4 min.
- **Total compute (sum across models):** 1.9 min.
- **Library versions:** torch 2.10.0+cu128, torchvision 0.25.0+cu128, transformers 5.0.0, pillow 11.3.0.
- **GPU:** Tesla T4.
- **Seed:** 42 (set on random, numpy, torch).
- **Re-run:** `python scripts/run_typographic_benchmark.py` then `python scripts/build_typographic_report.py` then `python scripts/build_gradient_report_pdf.py --input results/typographic/REPORT.md --output results/typographic/REPORT.pdf`. Per-model JSONs are the resumption unit — delete one to force its recomputation.